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RESEARCH ARTICLE

The impact of root systems on soil macropore abundance and soil infiltration capacity

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Abstract

Background and aims Soil infiltration influences the amount of precipitation entering the soil, which is particularly important in water-limited regions. Investigating the impact of root decay on soil porosity and infiltration rates is essential.

Method Natural grassland underwent a five-year film mulching treatment to suppress plant root growth, named grassland under film mulching (GFM). Simultaneously, natural recovery grassland (NRG) and farmland (FL) were selected for conducting in-situ infiltration experiments.

Results The diameter of roots significantly influences the formation of soil macropores ($p<0.05$). In the NRG soil layer less than 20 cm deep, the effective macroporosity with ø > 3 mm is significantly higher compared to other sites ($p<0.05$), which leads to enhanced infiltration capability. Upon reaching a soil depth of 20 cm, the stable infiltration rate of FL was 2.00 mm/min, showing no significant change with further increases in soil depth. The stable infiltration rate of GFM is measured at 6.95 mm/min, which is 5.7 times higher than that of NRG. The effective macroporosity of GFM (ø > 3 mm) increased by 20.6 times, with macropore connectivity reaching 20.79%, meanwhile that the effective macroporosity of NRG (ø > 3 mm) decreased by 93.67%.

Conclusion Although live roots contribute to the formation of macropores and infiltration capacity, the long and continuous biogenic macropores formed after root decay have a more pronounced effect on enhancing soil water infiltration capacity. This research endeavors to serve as a reference material for exploring the impact of grassland vegetation restoration on macropores and groundwater circulation.

Keywords Soil infiltration · Soil macropore · Dead roots · Root characteristics · Steady infiltration rate

Abbreviations

GFM	Grassland under film mulching
NRG	Natural recovery grassland
FL	Farmland
EM	Effective macroporosity
SM	Static macroporosity
RB	Root biomass
RL	Root length
RSA	Root surface area
ARD	Average root diameter
SIR	Steady infiltration rate
MC	Macropore connectivity

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Introduction

Soil moisture deficiency represents a critical issue that urgently needs to be addressed for ecological restoration in the semi-arid climate of the Loess Plateau (Fu et al. 2016). Infiltration, as the primary hydrological process converting precipitation into soil moisture, merits significant attention, notably in arid and semi-arid regions. Despite well-established recognition of soil porosity constraints on soil infiltration (Baets et al. 2011), the contribution of soil macropores to rainfall infiltration has been investigated (Beven and Germann 1982), however, uncertainties still exist, and it is difficult to quantify them (Beven and Binley 2014). Macropores significantly impact soil infiltration rates, especially during near-saturation infiltration (Scott et al. 1988). Investigating changes in soil pore structure due to root influence and soil infiltration is of profound significance in mitigating surface runoff and enhancing groundwater replenishment from rainfall, especially in arid and semi-arid ecological hydrology.

Numerous scholars have explored the mechanisms of soil macropores and their impact on infiltration. Nachabe MH (1995) observed, when using a tension infiltrometer to measure soil water conductivity on the soil surface, that the hydraulic conductivity of macropores was 3.6 times greater than the matrix of soil without macropores. Some researchers have identified soil macropores through dye tracing experiments (Mitchell et al. 1995; Wang and Zhang 2011), while others have quantified soil macropores by conducting CT scans on soil columns (Hu et al. 2015, 2018). The main causes of soil macropore formation can be categorized into three types: tubular pores created by soil fauna activity and decaying plant roots; natural cracks formed by physical processes such as wetting–drying and freeze–thaw cycles; and irregular pores between soil aggregates of varying particle sizes (Buttle and Leigh 1997). Among these, root channels and pores formed by soil fauna are typically tubular and highly connected, playing a crucial role in enhancing soil infiltration capacity. Researchers have also explored various relationships between different vegetation types and soil macropores (Gantzer and Anderson 2002; Zhou et al. 2008; Zhang et al. 2020). McCallum et al. (2004) found that perennial grasses left behind more macropores compared to annual grasses, and a significant network of mature plant roots contributes to the development of these macropores. Hagedorn and Bundt (2002b) found that bio-macropores formed by root systems in structured forest soils, as investigated through radionuclide studies, exhibited strong stability, capable of persisting for decades. These studies collectively draw similar conclusions, indicating that plant root systems are among the primary factors influencing soil macropore characteristics.

The impact of plant root systems on soil macropores manifests in three aspects: the growth activity of live roots creating gaps around the root vicinity; root channels formed by the decay of dead roots or root shrinkage; and live roots reoccupying the gaps formed by decomposed roots. Consequently, both live and dead roots can augment soil macroporosity and facilitate preferential flow occurrences (Zhang et al. 2020). On the one hand, soil texture also affects this process, studies have shown that soils with clay contents between 10 and 40% exhibit poorer aggregate stability compared to soils with higher clay contents, making it easier for dispersed aggregates to clog the pores (Bargués-Tobella et al. 2024; Ben-Hur et al. 1985). Conversely, soils with developed structure and finer texture tend to possess a greater abundance of macropores (Fischer et al. 2014; Tang et al. 2019). On the other hand, soil texture plays a crucial role in influencing root growth patterns (Stirzaker et al. 1996), which subsequently affects the formation of soil macropores. Specifically, plants tend to develop thicker or weaker, shorter roots in soils characterized by a dense structure and hard texture (Jin et al. 2015). In soils characterized by a soft and loose texture, root growth is less impeded, resulting in higher root length density in the topsoil layer (Lado et al. 2004).

The growth activities of the root system create larger pores within the soil (Bogner et al. 2012). When the root system grows in soil composed of loose small aggregates, the roots are able to displace surrounding soil particles from their growth path, thereby maintaining their growth trajectory, and influencing the formation of large pores (Ball et al. 2005; Bodner et al. 2014). In contrast, thick roots are more resistant to bending under mechanical pressure, thus exhibiting stronger capability to move soil particles (Clark et al. 2003; Gish and Jury 1983). Fine roots in soils with coarser texture tend to bend when encountering soil particles, growing along the gaps between soil particles, potentially leading to blockages (Bodner et al. 2014; Lu et al. 2020). Soil particles can only be displaced and soil macropores created in soils with finer texture (Tang et al. 2019). Cheng et al. (2011) suggested that active thick roots are unlikely to generate stable macropores; instead, root activity leads to soil contraction, resulting in the formation of fissures. The shrinkage of thick roots is crucial for fostering the development of macropores (Murielle et al. 2011), while the root channels formed as a result of root decay and decomposition constitute the primary source of these soil macropores (Newman et al. 2004; Ngao et al. 2007).

The current research concerning the relationship between soil macropores and root systems mostly focuses on analyzing root characteristics such as root biomass and active root traits, without considering the root channels left by dead roots. The consensus suggests that the quantity of root systems positively influences the number of soil macropores. However, the comprehensive elucidation of root system mechanisms remains incomplete, with analyses primarily addressing the impact of active roots on soil macropores. A minority of studies examining the comparison of effects of live and dead roots on infiltration and macropores primarily involve woody plants (Zhang et al. 2020). Research on grassland root systems in this regard is limited and primarily focused on field experiments (Mitchell et al. 1995; McCallum et al. 2004). However, grasslands, the primary vegetation type in arid and semi-arid regions, lack relevant field surveys.

Due to the lifespan of grassland plant root systems varies from a few days to three to five years (Strand et al. 2008), our study involved selecting a natural grassland covered with plastic film for five years, which inhibited the growth of its plant root system, designated as the grassland film mulching (GFM). In addition, a grassland that had been naturally restored for 40 years after farmland abandonment (NRG) and an agricultural land with corn cultivation (FL) were selected as the study objects. The aim is to establish the relationship between root development and mortality of different land-use types and the formation of soil macropores, as well as their impact on soil infiltration, through the analysis of root characteristics, macropore characteristics, and infiltration characteristics in these sample plots. This is intended to provide a reference for studies on grassland vegetation restoration in relation to macropore formation and groundwater circulation.

Materials and methods

Experimental site

The study was conducted in the Wangdonggou small watershed (35°12' ~ 35°16'N, 104°40' ~ 107°42'E), which is south of the Loess Plateau and located in Changwu County, Shaanxi Province, China (Fig. 1). It ranges in altitude from 940 to 1220 m. The climate is characterized as a warm temperate semi-humid continental monsoon, with an average annual temperature of 9.1°C and a frost-free period of 171 days. The average annual precipitation is 584 mm, exhibiting significant interannual variability. Precipitation distribution within the year is uneven, with low rainfall in spring and abundant rainfall in summer, mainly concentrated from July to September, accounting for over 55% of the total annual precipitation.

Predominant soil of the experimental site is a black loessial soil based on the Chinese soil classification and Calciustoll, according to the U.S.A. taxonomy, respectively. The collection of soil samples and plant root samples at the experimental site was divided into the following three areas: (1) NRG: The main plant species include Lespedeza (*Lespedeza bicolor*), Reed (*Phragmites australis*), Crested Wheatgrass (*Agropyron cristatum*), Gmelin's Wormwood (*Artemisia gmelinii*), with vegetation coverage of 95%. (2) GFM: Occasional presence of Crested Wheatgrass (*Agropyron cristatum*) and Edelweiss (*Leontopodium dedekensii*). (3) FL under manual tillage for comparison with NRG and GFM: The corn planted in the experimental plots was at the 3 leaves unfolded during the experiment.

Soil sampling and analysis

The soil samples were collected in April 2022. In each plot, soil samples from the soil layers of 0–10 cm, 10–20 cm, and 20–30 cm were collected using a soil auger and cutting rings (50.46 mm in diameter and 50 mm in height) from three points. Soil bulk density and initial moisture content were determined using the ring knife method. After air-drying, soil organic matter (SOM) was determined using the

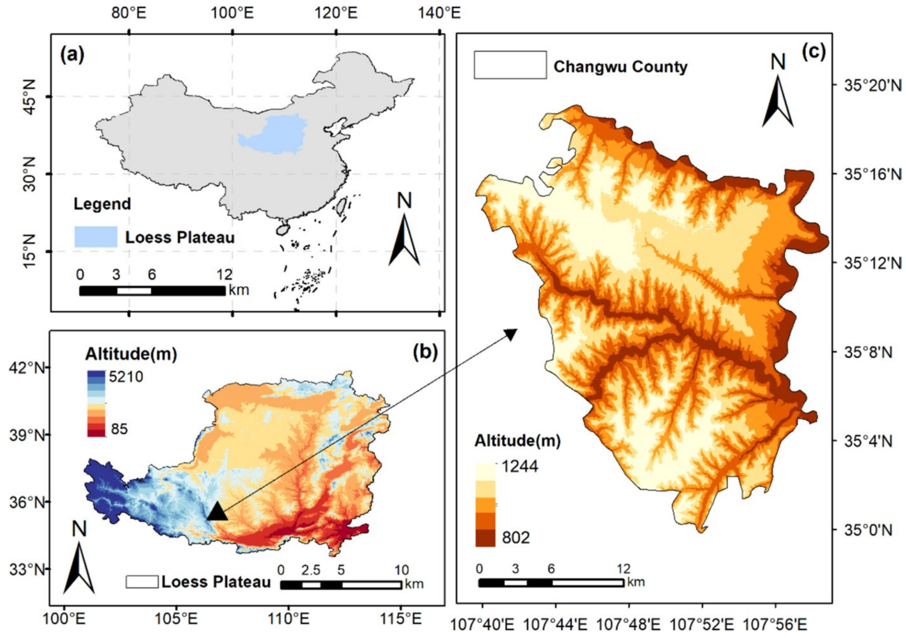


Fig. 1 Location of the Loess Plateau and Changwu County

K_2CrO_7 titration method, and soil particle size distribution was analyzed using a MasterSizer2000 laser particle size analyzer. The measured basic properties of the soil are shown in Table 1.

Root sampling and analysis

The root samples were collected by using a cutting ring with a diameter of 100 mm and a height of 63.7 mm. Three random sampling points were selected near the infiltration test location. Soil samples were collected from the soil layers of 0–10 cm, 10–20 cm, and 20–30 cm at the sampling points. After bringing the specimens to the laboratory, the samples were sieved through a 1mm mesh to clean them, extracting the roots from the soil. The root systems were scanned into images using an Epson Perfection V700/750 scanner, followed by the analysis of root morphology parameters from the images using WinRHIZO 2009 software. Upon completion of scanning, the vegetation root system was transferred into kraft bags and subjected to a 60°C oven until it attained a constant weight, for the determination of root biomass. The definition of macropores by Warner et al. (1989) was adopted, where pore sizes larger than ø > 1 mm were selected for classifying macropore sizes.

Table 1 The basic soil properties of plots in the 0–30 cm layer

Plots	Soil depth (cm)	Bulk Density (g/cm^3)	total porosity(%)	Soil separates			Organic Matter Content(%)
				Sand (%)	Silt (%)	Clay (%)	
NRG	0 ~ 10	1.11 ± 0.01	58.28 ± 0.50	48.67 ± 0.35	33.14 ± 0.91	18.19 ± 0.56	1.39 ± 0.01
	10 ~ 20	1.15 ± 0.02	56.73 ± 0.87	47.40 ± 1.36	32.06 ± 1.23	20.53 ± 0.38	0.96 ± 0.03
	20 ~ 30	1.24 ± 0.01	53.06 ± 0.24	47.12 ± 0.84	32.77 ± 0.86	20.10 ± 0.59	0.87 ± 0.01
GFM	0 ~ 10	1.04 ± 0.05	60.61 ± 2.06	49.17 ± 0.20	32.05 ± 0.30	18.78 ± 0.27	1.72 ± 0.01
	10 ~ 20	1.07 ± 0.05	59.76 ± 1.95	42.39 ± 0.23	34.53 ± 0.38	23.09 ± 0.14	1.20 ± 0.05
	20 ~ 30	1.13 ± 0.08	57.27 ± 2.83	46.36 ± 0.50	32.37 ± 0.24	21.27 ± 0.61	1.07 ± 0.06
FL	0 ~ 10	1.14 ± 0.01	57.08 ± 0.36	46.28 ± 1.09	30.23 ± 1.36	22.67 ± 0.31	1.17 ± 0.02
	10 ~ 20	1.19 ± 0.01	55.23 ± 0.31	46.68 ± 0.36	29.92 ± 0.39	21.72 ± 0.34	0.93 ± 0.09
	0 ~ 30	1.25 ± 0.01	52.78 ± 0.15	46.42 ± 0.92	31.26 ± 1.01	21.49 ± 0.40	0.89 ± 0.01

Soil macropores were classified into two categories based on infiltration experiments: 1mm < ø < 3mm, and ø > 3 mm (Seven and Germann 1981). Furthermore, coarse and fine roots are distinguished based on a root diameter of 1 mm, with thick roots further categorized into two types: medium- thick roots with diameters between 1–3 mm and very thick roots with diameters greater than 3 mm.

Determination and analysis of steady-state infiltration rate

Experiments were carried out from May 30th to June 15th, 2022. The disc permeameter (with a chassis diameter of 10 cm) was employed to measure soil infiltration rates at different soil depths: soil surfaces (0cm), after removing the upper 10cm layer (10cm), and after removing the upper 20cm layer (20cm). The following steps were performed in sequence: prior to each measurement, the water temperature was noted, the water head was adjusted (0 cm, −1 cm, and −3 cm), and the water level in the storage tube was recorded. Timing is started when the wetting front penetrates through the sand layer after the valve is opened. Throughout the infiltration process, data were recorded at 10-s intervals during the initial 1.5 min, followed by timing every 30 s from 1.5 to 3 min. After the initial 3-min period, data were recorded at 1-min intervals, with continuous recording until stable infiltration was reached. The procedure was replicated three times. The experimental protocol was guided by the relevant research conducted by Bodner (2014) and Nachabe MH (1995). According to Darcy's (H. Darcy) Law, the calculation formula for the steady-state infiltration rate, denoted as f_s, is as follows:

$$f_s = \frac{\Delta H \pi (D_2 \times 1/2)^2}{\Delta t \pi (D_1 \times 1/2)^2 C_T} \tag{1}$$

$$C_T = 0.7 + 0.03T \tag{2}$$

In the equation: f_s symbolizes the soil infiltration rate (mm/min) under standard water temperature conditions at 10°C; D_1 refers to the effective diameter of the disc permeameter base (cm); D_2 represents the diameter of the disc permeameter's storage tube (cm); Δt stands for the time interval (min); ΔH represents the difference in water storage tube readings during the time interval Δt (mm); C_T denotes the temperature correction coefficient; T denotes the average water temperature (°C) during a specific period.

Determination and analysis of soil macropore characteristics

The macropore calculation formula proposed by Watson and Luxmoore (1986) was used to compute the effective macropores in soil that participate in water transportation. This approach entails employing a disc permeameter to measure infiltration rates under designated pressure heads, thereby excluding specific pore sizes during the infiltration process. The macropore flow rates were obtained by measuring the infiltration rates multiple times at water heads of 0 cm, −1 cm, and −3 cm. The calculation formula is as follows:

$$N = \frac{8\mu[f(h_i) - f(h_{i+1})]}{\pi \rho g (r_m)^4}; i = 1,2,3,4 \tag{3}$$

$$\theta_m = N\pi (r_m)^2 = \frac{8\mu K_m}{\rho g (r_m)^2} \tag{4}$$

$$r_m = -\frac{2\sigma cos\beta}{\rho g h} \approx -\frac{0.15}{h} \tag{5}$$

In the equation: N represents the maximum number of effective macropores per unit area; μ denotes the viscosity coefficient of water (g·m^{-1}·s^{-1}); ρ denotes the density of water (g·m^{-3}); while g represents gravitational acceleration (m·s^{-2}); h_i corresponds to the respective water head; $f(h_i)$ and $f(h_{i+1})$ refer to the hydraulic conductivity at two corresponding pressure heads (m·s^{-1}); r_m signifies the minimum value set for the pore range (m); θ_m indicates macroporosity (%); σ represents the surface tension of water (g·s^{-2});β represents the contact angle between water and the pore wall (assumed to be 0). According to the capillary elevation equation (Watson and Luxmoore 1986), at h_i values of −1cm and −3cm, the associated equivalent pore diameters are 3mm and 1mm, hence, the quantities of macropores ø > 3mm、1 < ø < 3mm and ø < 1 can be calculated.

Static macropores are defined as pore structures in the soil with diameters falling within the size range of non-capillary macropores. However, they may not always contribute to water transport (Beven 1981). In this experiment, the static macroporosity was calculated using the soil water characteristic curve method. Samples, stratified into 0-10cm, 10-20cm, and 20-30cm layers using a cutting ring, were retrieved in their undisturbed state. These undisturbed soil samples were transported to the laboratory, fully saturated in water. The samples were centrifuged using a CR21 high-speed constant temperature centrifuge (manufactured by HITACHI, Japan) at 11 different set speeds until equilibrium was achieved. This process enabled the determination of soil volumetric water content at suction values of 0, 1, 10, 20, 40, 60, 80, 100, 200, 400, 600, and 800 kPa, thereby obtaining the soil moisture characteristic curve during soil desaturation. In this study, the Van-Genuchten model is utilized due to its high applicability and satisfactory model parameter fitting capabilities in order to fit the water characteristic curve (Lu et al. 2020). The model expression is as follows::

$$\theta = \theta_r + \frac{\theta_s - \theta_r}{\left[1+(\alpha h)^n\right]^m} \tag{6}$$

In the equation: θ(h) is the soil volumetric water content (cm^3 /cm^3); θ_s is the saturated water content (cm3 /cm3); θ_r is the soil residual water content (cm3 /cm3); h is the pressure head; α, m, n are model parameters, and $m = 1 - 1/n$.

The calculation of the equivalent pore diameter associated with static soil macropores employs the Rushton formula, $d = 3/p$, where p denotes the water column height (cm) of the applied suction (Koliji et al. 2006). The volume of water lost from the soil at corresponding suction values is calculated using the Van Genuchten model of the soil–water characteristic curve. This, in turn, enables the derivation of the macroporosity and the number of macropores with different equivalent pore diameters (Cameron and Buchan 2006).

Data processing

Data organization and management were performed utilizing Microsoft Excel 2019. Soil moisture characteristic curves were fitted using RETC software, and statistical analyses encompassing variance and correlation were executed via SPSS 22.

Results

Root characteristics

Figure 2(a) demonstrates a declining trend in root biomass for NRG, GFM, and FL as soil depth increases, with their roots predominantly concentrated within the 0-10cm soil layer. When soil depth increases from the 0–10 cm layer to the 10–20 cm layer, the root biomass of NRG decreases by 82.46% compared to the previous layer, and when it further increases from the 10–20 cm layer to the 20–30 cm layer, the decrease

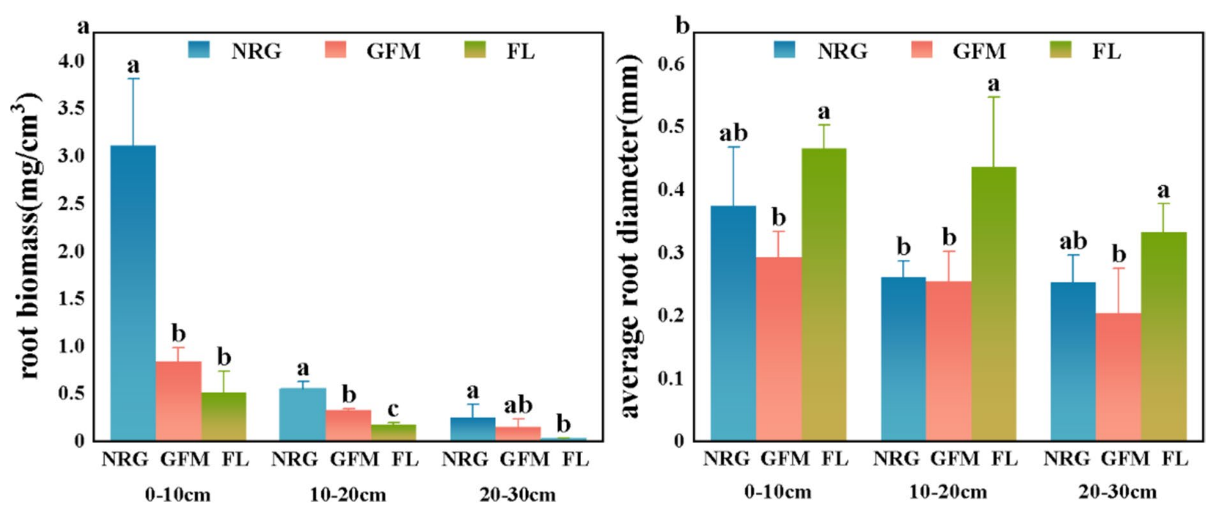


Fig. 2 The root biomass (**a**) and average root diameter (**b**) were determined across various land utilization types. Different letters represent significant differences ($p < 0.05$) among the three locations within the 0–10 cm, 10–20 cm, and 20–30 cm soil layers. The vertical bars represent the standard deviation

is 53.05%. Significant disparities exist in root biomass among various sampling sites within identical soil depth. Specifically, within the 0-10cm soil layer, NRG exhibits root biomass 3.7 times higher than GFM and 6.3 times higher than FL. FL displays significantly lower root biomass across varying soil layers compared to other sampling sites, which contrasts with FL's root mean diameter as indicated in Fig. 2b.

Figure 3 illustrates that the proportion of thick roots within the total root system across all sites is quite small, particularly the extremely thick roots. Additionally, thick roots are primarily concentrated within the 0-10cm soil layer. In all soil layers, NRG exhibits significantly higher values for total root length, total root surface area, and total root volume compared to GFM and FL. In the 0-10cm and 10-20cm layers, GFM displays a larger total root length compared to FL, while in the 20-30cm layer, FL's total root length slightly surpasses that of GFM (Fig. 3a). Additionally, the distribution of root surface area and root volume across different sites shows a similar trend. Despite constituting a minimal fraction of the overall root length, ranging from 2.1% to 5.1%, thick roots exhibit a proportionately higher presence in total root volume, spanning between 44.3% and 58.7%, owing to their larger diameter. In the aggregated root volume across distinct sites, a more pronounced exhibition of root diameter distribution discrepancies is discernible (Fig. 3c). The 0-10cm and 10-20cm layers of NRG exhibited a higher distribution of extremely thick roots with diameters > 3mm. FL exhibited a minor presence of extremely thick roots within the 10-20cm and 20-30cm layers, whereas GFM displayed a limited occurrence of such roots confined to the 0-10cm layer.

Soil infiltration

Figure 4 illustrates the variations in steady-state infiltration rates across different soil depths for NRG, GFM, and FL, corresponding to changes in matric suction (i.e., negative water head) at the soil surface. With a matric potential of 0, the stable infiltration rates among the different locations in the surface soil layer (0 cm) exhibited significant differences: NRG > FL > GFM ($p < 0.05$). The rates were 2.61 mm/min, 2.06 mm/min, and 1.43 mm/min, respectively (Fig. 4a). At a soil depth of 10 cm, NRG showed the highest stable infiltration rate compared to the other locations (Fig. 4b). At a soil depth of 20cm, the measured steady-state infiltration rates for NRG, GFM, and FL are 1.22 mm/min, 6.95 mm/min, and 2.00 mm/min, respectively. The steady-state infiltration rate of GFM is 5.7 times that of NRG and 3.5 times that of FL.

With the increase of matric suction, the decrease in pore size and quantity participating in water conduction resulted in an overall decreasing trend in stable infiltration rates. Yet the rates and extents of decline vary, reflecting the quantity and size of macropores involved in water infiltration in the soil at each location. To enhance the explanation of soil infiltration capacity and the role of underground roots and macropores in soil infiltration, infiltration tests were performed on the soil surface at depths of 10 cm and 20 cm. At soil depths of 0 cm and 10 cm, the stable infiltration rate of NRG shows a rapid decline at a water head of −1 cm, with reduction rates of 77.4% and 71.1%, respectively. This indicates the involvement of soil macropores larger than 3 mm in water infiltration. The stable infiltration rate of FL exhibited the largest decrease, ranging from 23.2% to 28.2%, as the water head decreased from −1 cm to −3 cm, during which pores with a diameter larger than 1 mm participated in the infiltration process. Throughout the process of transitioning from 0 cm to −3 cm of pressure head, the stable infiltration rate of GFM showed minimal variation. Upon reaching a soil depth of 20cm, distinct alterations in the steady-state infiltration rate curve features are evident across all sample sites. The rapid decrease in the stable infiltration rate of GFM as the water head decreased from 0 cm to −1 cm suggests a larger number of soil macropores with diameters greater than 3 mm. The stable infiltration rate of NRG showed a trend of initially increasing and then decreasing with the increasing soil depth across different soil layers. The stable infiltration rate of NRG at the soil surface of the 10 cm layer continued to be significantly elevated compared to other locations, the stable infiltration rate of GFM exhibited a minor increase, nearing that of FL (Fig. 4b). At the soil surface of the 20 cm layer, the stable infiltration rate of NRG was observed to be lower than that of FL, whereas conversely, GFM demonstrated a notably higher stable infiltration rate compared to other locations ($p < 0.05$) (Fig. 4c). The steady-state infiltration rate of FL shows no significant change with increasing soil depth, whereas that of GFM consistently increases.

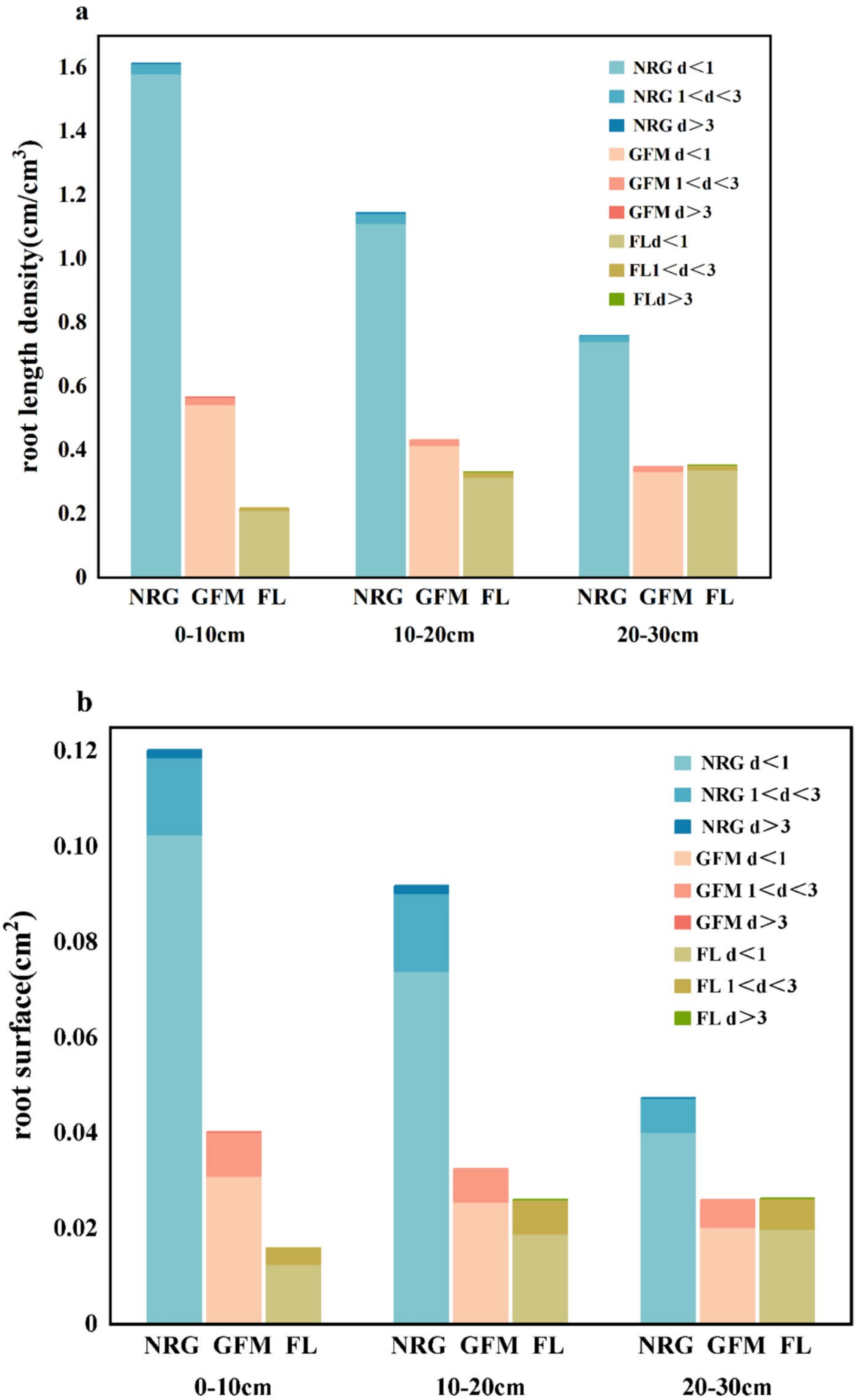
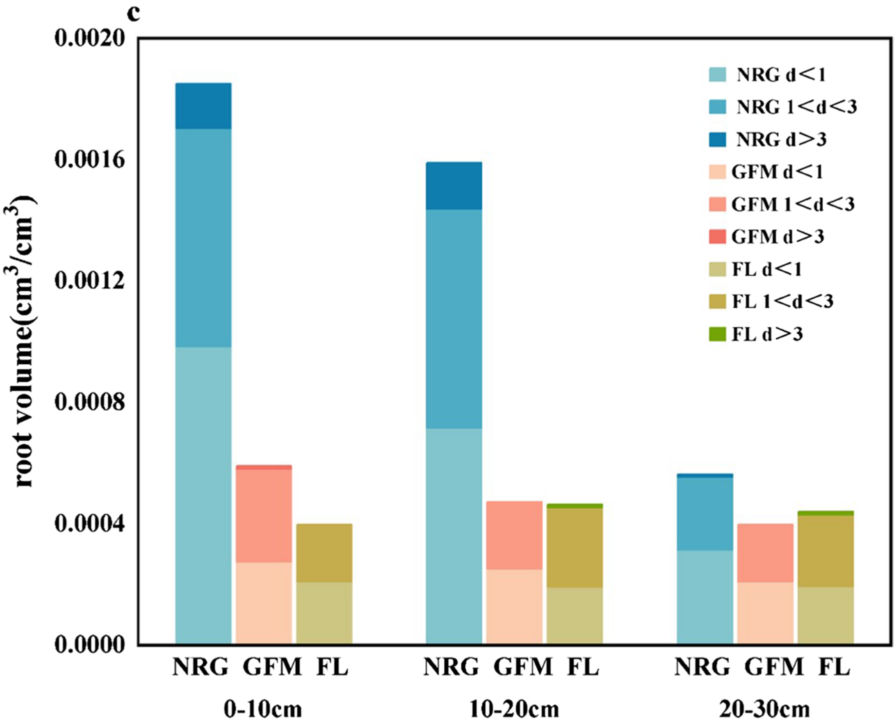


Fig. 3 The roots of the three types of plots were classified based on root diameter (**d**). The root length (**a**), root surface area (**b**), and root volume (**c**) corresponding to roots with d < 1mm, 1mm < d < 3mm, and d > 3mm were presented

Fig. 3 (continued)



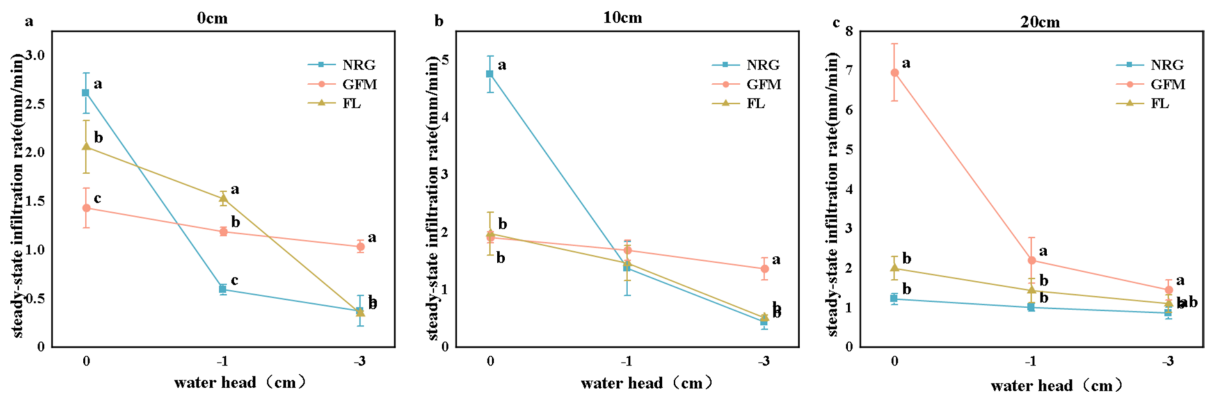


Fig. 4 The steady-state infiltration rates were measured across three distinct plots, subjected to varying water head pressures, and assessed at different soil layers (0cm, 10cm, 20cm). Different superscript letters indicate significant differences in the stable infiltration rates between different plots under the same water head ($p < 0.05$). The vertical bars represent the standard deviation

Soil macropores distribution

Effective macropore distribution

This study adopts the definition of effective macropores from the perspective of water flow dynamics by Beven and Germann (1982), that is, the soil structure in which non-equilibrium conduction can occur. Table 2 illustrates the distribution of effective macroporosity (EM) in soil participating in water transport, it can be observed that the macropore which is the minute fraction of soil volume contributes greatly to hydraulic conductivity, it can even be as high as 90.8%. At both the soil surfaces of 0cm and 10cm, the

Table 2 The distribution of effective macropores in different soil layers across various plots

Depth (cm)	Plot	Pore size class (mm)	Maximum number of effective macropore per unit area (individual·m^{-2})	Maximum effective macroporsity (%)	Accumulated macropore impact on the total flow rate (%)
0	NRG	> 3	2.26	0.01599	77.38
		1–3	19.99	0.01569	85.82
	GFM	> 3	0.11	0.00081	8.19
		1–3	9.86	0.00077	16.93
	FL	> 3	0.59	0.00042	25.8
		1–3	107.19	0.08414	83.29
10	NRG	> 3	3.78	0.02668	71.11
		1–3	84.71	0.06649	90.8
	GFM	> 3	0.18	0.00013	8.44
		1–3	34.58	0.02715	28.5
	FL	> 3	0.57	0.00040	25.92
		1–3	86.59	0.06797	74.3
20	NRG	> 3	0.51	0.00036	37.68
		1–3	35.34	0.02774	69.66
	GFM	> 3	5.32	0.03757	68.36
		1–3	68.27	0.05359	79.2
	FL	> 3	0.53	0.00038	23.78
		1–3	38.21	0.03000	44.89

Different superscript letters indicate significant differences between different plots in the same soil layer ($p < 0.05$)

presence of NRG ø > 3 mm effective macropores is significantly higher than in other locations ($p < 0.05$). At the soil surface (0cm), the quantity of effective macropores of NRG ø > 3 mm and those ranging from 1 < ø < 3mm are documented as 2.26 and 19.99 pores per square meter, respectively. In the equivalent layer, the quantity of effective macropores with diameters > 3mm in FL is significantly lower than that observed in NRG ($p < 0.05$). However, the quantity of effective macropores ranging in size from 1 to 3 mm is significantly higher than in other sample sites, with a count of 107.19 per square meter ($p < 0.05$). The effective macroporosity of GFM is significantly lower than that of the other two sample sites ($p < 0.05$). With soil depth reaching 10cm, both NRG and GFM show an increase in effective macroporosity, while FL exhibits a slight decrease. At a soil depth of 20cm, a significant shift in soil effective macroporosity is evident. A significant decrease of 93.67% was observed in the effective macroporosity of NRG for pores exceeding 3 mm in diameter, while the effective macroporosity of pores ranging from 1 to 3 mm also diminished by 36.56%, resulting in values lower than those for FL. The effective macroporosity of NRG with ø > 3 mm is significantly lower than that of GFM ($p < 0.05$). When the soil depth increases to 20 cm, the effective macroporosity of GFM increases by 20.6 times. Meanwhile, GFM's effective macroporosity distinct increases, notably reaching 5.32 per square meter for > 3 mm macropores, with its 1-3mm effective macroporosity considerably higher than that of NFG and FL. The effective macroporosity of FL > 3mm shows almost no change across different soil layers, consistently ranging between 0.00038–0.00042.

Static macropore distribution and macropore connectivity

Figure 5 illustrates the soil water characteristic curves for the three sample sites, NRG, GFM, and FL, at different soil depths (0–10 cm, 10–20 cm, 20–30 cm). Fitting of the measured data was performed using the VG model. The results of the fitting parameters are presented in Table 3, where the goodness-of-fit R^2 values for all sample sites are above 0.98. In the

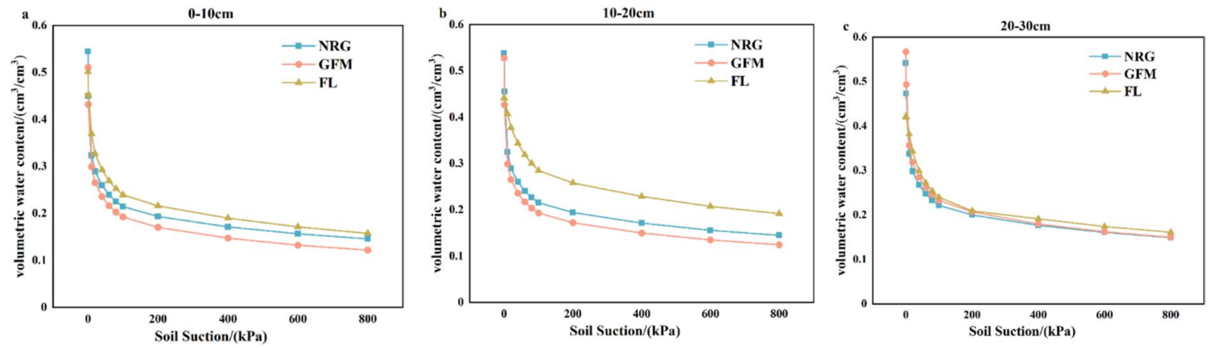


Fig. 5 Soil water characteristic curves for the three types of sample sites at different soil layers (0cm, 10cm, 20cm)

Table 3 Curve-fitting parameters for the soil moisture characteristics of the three plots

Plots	Soil depth (cm)	θ_r	θ_s	α	n	m	R^2
NRG	0 ~ 10	0.0644	0.5434	0.0104	1.5105	0.3379	0.9981
	10 ~ 20	0.0682	0.5367	0.0109	1.5031	0.3347	0.9991
	20 ~ 30	0.0652	0.5408	0.0111	1.5054	0.3357	0.9995
GFM	0 ~ 10	0.0671	0.5094	0.0106	1.4984	0.3326	0.9983
	10 ~ 20	0.0755	0.5266	0.0097	1.5134	0.3392	0.9981
	20 ~ 30	0.0700	0.5640	0.0107	1.5034	0.3343	0.9981
FL	0 ~ 10	0.0739	0.4908	0.0117	1.4851	0.3236	0.9955
	10 ~ 20	0.0725	0.4429	0.0120	1.4837	0.3260	0.9987
	20 ~ 30	0.0692	0.4219	0.0117	1.4922	0.3298	0.9981

water characteristic curve, as water suction increases, the moisture in the larger pores of the soil matrix is released first, followed by the release of water in pores of successively smaller diameters. When the soil water suction h is greater than 100 kPa, the volumetric water content in the 0–10 cm and 10–20 cm layers for the various sample sites follows the order FL > NRG > GFM. This indicates that the FL soil contains a higher proportion of small pores (Fig. 5).

Static macroporosity (SM) refers to the maximum pore space existing in the soil, representing a static pore structure that does not necessarily participate in water transport (Beven and Germann 1982). These macropores may contain dead-end spaces, while another portion of the macropores may be continuous, facilitating rapid water transport. The connectivity of macropores is measured by the ratio of effective macroporosity participating in water transport to static macroporosity (Bodhinayake and Cheng Si 2004). Figure 6 shows the EM and SM of different soil layers in various plots, as well as macropore connectivity. The trends of static macropores with ø > 3 mm and 1 < ø < 3 mm are consistent across different plots, but there are significant differences in static macropores between different plots. In the 0–10 cm layer, the soil moisture (SM) of FL is greater than that of NRG and GFM. The static macroporosity of NRG shows an increasing trend with soil depth, while the trend for FL is opposite. GFM exhibits a trend of initial decrease followed by an increase. In the 10–20 cm layer, the soil moisture (SM) of GFM is significantly lower than that of NRG and FL ($p < 0.05$). As soil depth increases to the 20–30 cm layer, the static macroporosity across all sample sites becomes similar, with no significant differences observed. The water characteristic curves for the three sample sites also tend to converge (Fig. 5). For the NRG soil, the connectivity of macropores with diameters ø > 3 mm was measured to be 9.77% and 15.11% in the 0–10 cm and 10–20 cm soil layers, respectively. In the 20–30 cm layer, this connectivity decreased significantly to 0.92%, indicating a reduction of 93.90% (Fig. 6a). For the GFM soil, the connectivity of macropores with diameters ø > 3 mm in the first two layers is low, with

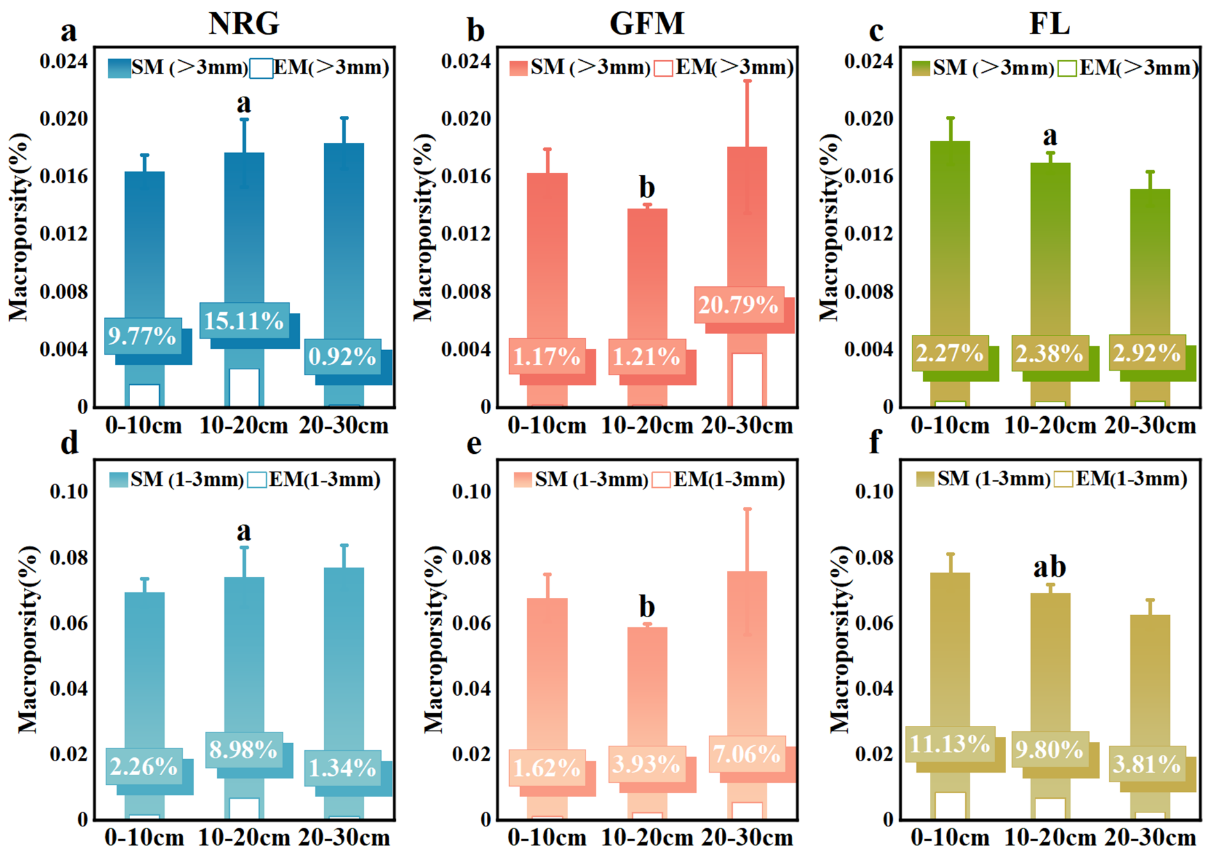


Fig. 6 Distribution of static macropores, effective macropores, and their connectivity in different soil layers of three sample plots. **a-c** showcase macropores with diameters exceeding 3 mm, whereas **d-f** illustrate micropores with diameters ranging from 1 to 3 mm, depicting their distribution, effectiveness, and connectivity across different soil layers of three sample plots. The vertical bars represent the standard deviation.SM denotes the static macroporosity; EM denotes the effective macroporosity. Histograms followed by a different superscript letter are significantly different ($p < 0.05$)

a range of only 1.17% to 1.21%. Notably, the connectivity in the 20 cm layer is the highest observed among all sample sites, attaining a value of 20.79% (Fig. 6b). The connectivity of macropores with diameters ø > 3 mm in the FL sample site is relatively low and exhibits minimal variation across different soil layers, with values of 2.27%, 2.38%, and 2.92%, respectively (Fig. 6c). However, the connectivity of macropores with diameters in the range of 1 < ø < 3 mm is notably higher than that of macropores with diameters ø > 3 mm (Fig. 6f).

Relationship analysis of soil pore size distribution and soil infiltration

As shown in Fig. 7, SIR displays a highly significant positive correlation with EM and MC for pores with diameters ø > 3 mm ($p < 0.01$). It also exhibits significant positive correlations with EM and MC for pores with diameters in the range of 1 < ø < 3 mm, but shows a significant negative correlation with mean root diameter ($p < 0.05$). The EM of pores with diameters ø > 3 mm is mainly affected by macropore connectivity and displays a significant positive correlation with MC for pores of the same size range ($p < 0.01$). Additionally, it exhibits a significant negative correlation with mean root diameter ($p < 0.05$). The EM of pores with diameters between 1 and 3 mm is negatively correlated with total root length density, root length density of roots with diameters less than 1 mm, and root surface area density ($p < 0.05$). In contrast, MC of pores within this size range displays a significant positive correlation with mean root diameter ($p < 0.05$).

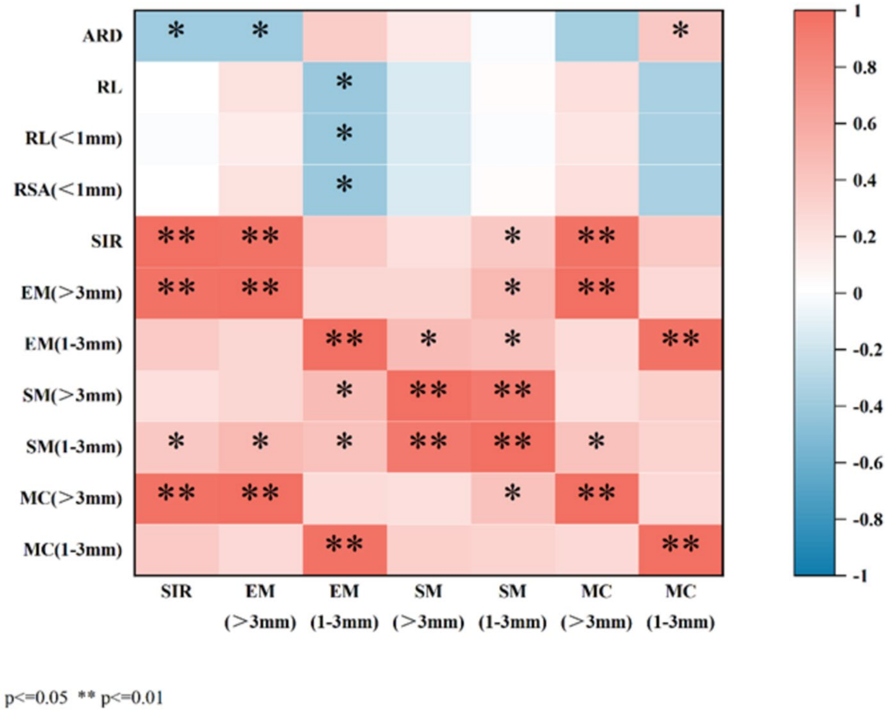


Fig. 7 Correlation between stable infiltration rate and characteristics of macropores and roots. ARD is average root diameter; RL is root length; RSA is root surface area; SIR is stable infiltration rate at zero negative pressure; MC is macropore connectivity. * was significantly correlated at the level of 0.05, and ** was significantly correlated at the level of 0.01

Discussion

The influence of root system variations on soil macropores

During the process of vegetation restoration, the growth of underground root systems can penetrate the soil (Dexter 1987; Whalley et al. 2004, 2005). Subsequently, the roots undergo decay and decomposition, leaving long and continuous biological pores (Ball et al. 2005; Cresswell and Kirkegaard 1995; Ni et al. 2018). Changes in root system characteristics during this process, such as root diameter, root length, and root biomass (Técher and Berthier 2023), as well as root exudates and surrounding microorganisms (Ball et al. 2005; Feeney et al. 2006; Whalley et al. 2005), also affect soil structure and the formation of soil macropores, thereby influencing the generation of preferential flow (Lange et al. 2009). Distinguishing itself from prior research on plant roots and soil macropores, this study distinguishes between root systems in natural grassland and mulched grassland, aiming to elucidate how changes in root characteristics impact soil macropore quantity and consequent alterations in soil infiltration capacity. Following the mulching treatment applied to natural grassland, GFM displayed a significant decrease in root biomass compared to NRG ($p<0.05$), alongside a slight decrease in average root diameter (Fig. 2). This treatment also led to marked reductions in total root length, total surface area, total root volume, and various other root characteristics (Fig. 3). The Fig. 3(c) distinctly illustrates that NRG exhibits a greater abundance of thick roots, including both those with diameters > 3 mm and those ranging from 1–3 mm, surpassing GFM. These findings indicate a substantial reduction in root biomass and thick root quantity in GFM due to mulching treatment, signifying a significant shrinkage and even decay of its underground root system.

Many studies suggest a close positive correlation between root traits in vegetation recovery and soil macroporosity (Bodner et al. 2014; Clark et al. 2003; Hu et al. 2015; Yunusa and Newton 2003). Researchers believe that thicker root systems have an easier time penetrating soil and are less prone to bending when encountering mechanical resistance, channels formed around thick roots exhibit greater stability, facilitating water transport. The results of this experiment showed a significant positive correlation between MC for pores with diameters between 1 and 3 mm and mean root diameter ($p<0.05$).

This suggests that roots with larger diameters facilitate the connectivity of macropores in the $1<ø<3$ mm range (Fig. 7). The results of various studies have shown that the intercalation of thick root systems within the soil leads to the formation of pores. (Clark et al. 2003; Holtham et al. 2007; Whalley et al. 2005). Nevertheless, on the other hand, such root activity can disturb the established larger pores, and in conjunction with the action of root exudates, it may conversely result in some macropores becoming discontinuous and losing their capacity for water transmission. (Whalley et al. 2004, 2005). As shown in Fig. 7, there are significant negative correlations observed between EM, SIR for pores with diameters $ø>3$ mm, and mean root diameter. Meanwhile unlike most researchers' experiments, the root system characteristics of GFM in this experimental design represent roots that have undergone atrophy and death. The decrease in root biomass and average root diameter indicates the formation of a relatively large number of root channels underground. The shrinkage or decay of thick roots can create biological macropores in the soil, and these long (Murielle et al. 2011), continuous root channels are one of the main reasons for the formation of effective macropores (Beven and Germann 1982; Luo et al. 2010; Shougrakpam et al. 2010). Relevant studies indicate that fine roots typically cannot form macropore channels, but instead tend to penetrate into preexisting pores (Bodner et al. 2014; Chen and Weil 2010; Lu et al. 2020). Figure 7 illustrates that both root length density and root surface area density of fine roots with diameters <1 mm exhibit significant negative correlations with EM for pores with diameters in the range of $1<ø<3$ mm ($p<0.05$), indicating that the presence of fine roots can disrupt soil macropores. Therefore, when the soil depth increases from 0 to 10 cm, the root biomass of NRG decreases by 82.46% (Fig. 2a), but the SM also increased by 8.21% accordingly (Fig. 6a, d). When the soil depth increases to 20 cm, the root biomass in NRG soil experiences a 53.05% decrease compared to the 10–20 cm layer. For pores with diameters $ø>3$ mm, SM increased by 3.73%, while EM decreased by 93.67%. Comparing the changes in effective macroporosity and connectivity in the same soil layer of other plots, it can be observed that the deep roots in NRG not only fail to generate additional pores but also interfere with and block the existing effective macropores.

The influence of roots and soil macropores on soil infiltration

Soil stable infiltration rate is an important indicator reflecting soil infiltration characteristics. As shown in Fig. 4a, under zero head pressure, the stable infiltration rate of NRG at the 0 cm layer is significantly higher than that of FL, while the stable infiltration rate of GFM is significantly lower than that of FL ($p<0.05$). This suggests that, in near-saturation conditions, which mimic natural rainfall environments, the infiltration capacity of NRG is significantly stronger than that of GFM and FL. The occurrence of such differences is related to the size and distribution of pores involved in water conduction in the soil, especially the effective macropores that can generate preferential flow (Bodhinayake and Cheng Si 2004).

There is no significant difference in the stable infiltration rate of FL among different soil layers (Fig. 4). This is related to the different land use patterns of FL, which is a cultivated cornfield. Due to agricultural machinery tillage, the static macroporosity in the 0–10 cm layer of the FL site was higher than that of the other sample sites (Fig. 6). Nevertheless, Fig. 4a indicates that the higher static macroporosity with pore diameters exceeding 3 mm ($ø>3$ mm) in the FL site did not lead to an enhancement of soil infiltration capacity when compared to the other sites. The soil infiltration rate is determined by the number of effective macropores, particularly those with diameters greater than 3 mm. However, in agricultural land where the soil is disturbed by human activities such as tillage, it is difficult for connected effective macropores larger than 3 mm in diameter to develop. In the case of FL, only a limited number of such pores are evenly distributed among different soil layers, and their connectivity significantly lags behind that of naturally occurring macropores.

The effective macropores of NRG exhibit the highest contribution rate at the 0 cm and 10 cm soil surfaces, particularly reaching a contribution rate of 90.8% in the 10 cm soil layer (Table 2). This indicates that soils in naturally restored grasslands possess excellent permeability and porosity, Soil macropores facilitate rapid water infiltration and provide an optimal growth environment for root systems. During natural recovery, NRG has developed a relatively large number of effective macropores with $ø>3$ mm, primarily concentrated in the top two soil layers. This phenomenon is related to well-developed plant root systems (Hu et al. 2018; Lu et al. 2020; McCallum et al. 2004). Generally speaking, the underground root system of natural grassland is mainly concentrated in the soil above a depth of 20 cm (Alberti and Cey 2011; Li et al. 2011). The total root volume of NRG also shows a significant decrease after the soil depth exceeds 20 cm (Fig. 3h). Related researchers have revealed that a well-developed plant root network facilitates the development of macropores (Mawodza et al. 2020). Furthermore, these macropores, formed by such roots, exhibit remarkable stability and can persist for extended periods of time (Bogner et al. 2012; Hagedorn and Bundt 2002a), facilitating soil water infiltration. However, the infiltration rate of NRG at the 20 cm soil surface decreased by 74.31%, which is related to a 93.67% reduction in its effective macroporosity (ø > 3 mm) and also associated with the growth pattern of the root system. The NRG possesses a considerable number of roots that penetrate deeply into the soil, and the roots growing in the deeper soil layers preferentially develop within pre-existing macropores (White and Kirkegaard 2010; Chen and Weil 2010; Giuliani et al. 2024), and other live roots may also occupy and fill macropores, resulting in their blockage and impeding water movement (Gish and Jury 1983). This behavior of roots growing into pre-existing pores also disrupts macropore connectivity, causing them to split into smaller pores (Scanlan 2009).

GFM represents a grassland that has undergone mulching treatment, where a significant portion of its underground root system has atrophied and died, leaving numerous root channels behind. In the absence of root growth disturbance, macropores in the lower soil layers are better preserved, except the upper soil layers, which are more susceptible to external stress. (Cheng et al. 2011). Consequently, at the 20 cm soil layer, GFM exhibits strong infiltration capacity due to the formation of long and continuous pores created by dead roots, which serve as channels for rapid water migration. (Mitchell et al. 1995). The infiltration rate of the top two layers in GFM is lower than FL, and this is associated with substantial wilting of its underground roots under the influence of five years of covering with mulch. As the root systems, which serve to support soil structure, are lost over an extended period, the original macropores in the soil also collapse. Figure 6(b) indicates that, despite the extremely low effective macroporosity of GFM in the soil surface layer, its static macroporosity is still comparatively high. A substantial number of macropores are left behind after root system death, however, the connectivity of these macropores cannot be sustained. Rainfall during mulching accumulates above the mulch film, unable to infiltrate, compacting the soil surface. Soil compaction causes a severe disruption to the topsoil pore structure (Chen and Weil 2010). The effect of this disruption decreases with greater soil depth, when reaching a depth of 10 cm, GFM exhibits a saturated hydraulic conductivity that closely approximates FL (Fig. 4b). The results indicate that with increasing soil depth, the influence of surface disturbances on the macropores within the soil decreases. Below 20 cm, eliminating this interfering factor, there is a notable abundance of biological macropores in the soil resulting from the atrophy and death of roots. These pores are well-preserved in the deeper soil layers, forming vertically stable and effective root channels. This allows for a connectivity of 25.29% for macropores with ø > 3 mm at depths of 20–30 cm (Fig. 6b). The infiltration capacity of GFM is stronger, with its stable infiltration rate being 5.7 times that of NRG. Many researchers have reached similar conclusions that most root channels and preferential flow channels are formed after the death of roots (Lu et al. 2020; Técher and Berthier 2023).

In the NRG soil above 20 cm, not only does it contain a large number of extensive roots, but also accumulated dead root channels from years of restoration, resulting in a significantly higher stable infiltration rate compared to other plots. Below 20 cm, roots tend to preferentially grow through macropores, which offer the path of least resistance (Giuliani et al. 2024; Gao et al. 2016), thereby blocking the macropores and affecting their infiltration capacity. In the view of Gish (1983) pre-existing root channels and active roots probably offset each other in field conditions. Root channels formed after plant root decay are crucial in the pathways of macropore flow, aligning with the findings of this study. Some researchers hold contrary views; Zhang et al. noted a larger influence of live roots on infiltration rate, whereas the impact of deceased roots was greater on soil water retention capacity (Zhang et al. 2020). The experimental results indicate that compared to NRG, GFM shows a significant increase in soil static macroporosity after root decay and decomposition (Fig. 6), favoring soil water retention. The support provided by live roots ensures stability in soil pore structure, facilitating water transport. Therefore, both live and deceased roots contribute to the formation of macropores, potentially increasing preferential flow occurrences (Zhang et al. 2020). However, the enhancement effect of root channels left after root death on infiltration is much greater than the impact of living roots on macropore formation. Some researchers have studied macropores and infiltration under different vegetation covers and found a correlation between vegetation density and infiltration capacity (Shougrakpam et al. 2010). However, as the duration of vegetation restoration increases, particularly in ecologically fragile regions such as the Loess Plateau, the soil's capacity to support vegetation becomes constrained. Moreover, there exists a mutual constraint between soil moisture and vegetation (Fu et al. 2016; Mei et al. 2018). Following a certain period of vegetation restoration, vegetation coverage tends to stabilize or even decline (Jin et al. 2022). Neveertheless, even if the aboveground biomass no longer increases, the subterranean root system persists in growth, mortality, and decomposition, consequently augmenting the organic carbon content within the soil (Qiao et al. 2019), and leaves a large number of root channels. The impact of roots on soil macropores and infiltration is not solely attributed to their growth activities; rather, it also encompasses the root channels formed after death and decay, which constitutes a crucial factor. This elucidates why the infiltration capacity increases with the prolonged duration of vegetation restoration (Qiu et al. 2022).

Conclusion

During the process of natural recovery, natural grasslands develop a rich root system, creating a significant number of macropores, with superior infiltration capability compared to cultivated agricultural land.

Following the application of mulch on natural grasslands, the grass root systems gradually shrink and die off, resulting in reduced root diameter and biomass. The diameter of roots significantly impacts the formation of soil macropores. The activity of coarser root systems is prone to disrupting well-developed effective macropores with diameters greater than 3 mm (ø > 3 mm) in the soil. After the contraction of thick roots, there is a conducive environment for the formation of effective macropores, enhancing water infiltration. Upon reaching a soil depth of 20 cm, away from the external compression experienced by the soil surface, the macropore connectivity of GFM reached 20.79%. The effective macroporosity of GFM with ø > 3 mm increased by 20.6 times, its stable infiltration rate is 5.7 times that of NRG. On the contrary, the effective macroporosity of NRG (ø > 3 mm) decreased by 93.67%. In the top 20 cm layer of NRG soil, the effective macroporosity of roots with a diameter exceeding 3 mm is notably higher than that of other plots, consequently leading to a stronger infiltration capacity in comparison to other plots. However, in the deeper soil layers, root growth may block large macropores, thus affecting their infiltration capacity. The enhancement effect on soil water infiltration capacity from the long and continuous biogenic macropores formed after root death far outweighs the influence of live roots on macropore formation.

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Data availability All data generated or analyzed during this study are included in this published article.

Declarations

Competing interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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